Exam AWS Certified Developer - Associate DVA-C02 All Questions

View all questions & answers for the AWS Certified Developer - Associate DVA-C02 exam

Go to Exam

EXAM AWS CERTIFIED DEVELOPER - ASSOCIATE DVA-C02 TOPIC 1 QUESTIO...

Exam question from Amazon's AWS Certified Developer - Associate DVA-C02
Question #: 119
Topic #: 1
[All AWS Certified Developer - Associate DVA-C02 Questions]

A real-time messaging application uses Amazon API Gateway WebSocket APIs with backend HTTP service. A developer needs to build a feature in the application to identify a client that keeps connecting to and disconnecting from the WebSocket connection. The developer also needs the ability to remove the client.

Which combination of changes should the developer make to the application to meet these requirements? (Choose two.)

A. Switch to HTTP APIs in the backend service.

B. Switch to REST APIs in the backend service.

C. Use the callback URL to disconnect the client from the backend service.

D. Add code to track the client status in Amazon ElastiCache in the backend service.

E. Implement $connect and $disconnect routes in the backend service.

Show Suggested Answer

---

by MrTee at *April 22, 2023, 11:24 p.m.*

Disclaimers:
- ExamTopics website is **not** related to, affiliated with, endorsed or authorized by Amazon.

- Trademarks, certification & product names are used for reference only and belong to Amazon.

Comments

MrTee Highly Voted 2 years, 11 months ago

Selected Answer: DE

Option D because by storing the client status in the cache, the backend service can quickly access the client status data without the need to query the database or perform other time-consuming operations.
Option E. Implement $connect and $disconnect routes in the backend service: $connect and $disconnect are the reserved routes in WebSocket APIs, which are automatically called by API Gateway whenever a client connects or disconnects from the WebSocket. By implementing these routes in the backend service, the developer can track and manage the client status, including identifying and removing the client when needed.

upvoted 22 times

akmv2 2 years ago

How is D viable when the question doesn't mention Elasticache? You're making an assumption that adding a service/configuration is what's being asked

upvoted 1 times

robotgeek 10 months ago

You guys need to stop pasting ChatGPT answers and specially upvoting them without any reference link

upvoted 2 times

catcatpunch Highly Voted 2 years, 10 months ago

Selected Answer: CE

C =>
https://docs.aws.amazon.com/ko_kr/apigateway/latest/developerguide/apigateway-how-to-call-websocket-api-connections.html

E =>
https://docs.aws.amazon.com/ko_kr/apigateway/latest/developerguide/apigateway-websocket-api-route-keys-connect-disconnect.html

upvoted 10 times

spandyda Most Recent 3 months, 1 week ago

Selected Answer: CE

CE is the answer.

upvoted 1 times

Wardove 8 months, 3 weeks ago

Selected Answer: CE

Although - option D is also correct, I would choose C and E
1) what we know 100% - we can disconnect uses using Callback url reference:
https://docs.aws.amazon.com/apigateway/latest/developerguide/apigateway-how-to-call-websocket-api.html
2) we definitely need to implement $connect and $disconnect routes (to identify during runtime)
3) we cannot technically track a user spamming connections if we don't

pass this info logically to each invocation - and technically without persistence this is a mission impossible

upvoted 1 times

mooncake1 1 year, 2 months ago

Selected Answer: CE

DE does not make the developer disconnect

upvoted 1 times

mooncake1 1 year, 2 months ago

Can't believe DE is 50%.. this q is not even hard

upvoted 1 times

Arad 1 year, 2 months ago

Selected Answer: CE

CE is correct.

upvoted 1 times

sumanshu 1 year, 3 months ago

Selected Answer: DE

A) Eliminated - Switching to HTTP APIs would eliminate the WebSocket capability entirely,
B) Eliminated - REST APIs are also unsuitable for WebSocket connections, as they are designed for request-response interactions rather than persistent, real-time communication.

Both C&E and D&E - seems correct

C) - API Gateway provides a callback URL that your backend service can use to invoke the POST /connections/{connectionId} API with the disconnect command.

D)

E) Correct - The $connect route triggers when a client establishes a WebSocket connection. The $disconnect route triggers when a client disconnects from the WebSocket

upvoted 1 times

sumanshu 1 year, 3 months ago

C) Callback URL allows you to explicitly disconnect clients.

upvoted 1 times

sumanshu 1 year, 3 months ago

The two requirements in the question are:

Identify a client that keeps connecting and disconnecting:

This requires tracking client connections and disconnections over time.
To achieve this, the backend needs a mechanism to log or store connection events for each client, which can then be analyzed to identify repeated connection patterns.
Remove the client:

Once the client is identified as problematic (e.g., frequent

connects/disconnects), the backend needs to remove the client. This is achieved using the WebSocket callback URL to explicitly disconnect the client.

upvoted 1 times

sumanshu 1 year, 3 months ago

D is critical for tracking client activity.
E is required to capture connection and disconnection events.
C provides the functionality to disconnect a client once identified.
Thus, the correct combination is C, D, and E.

upvoted 1 times

sumanshu 1 year, 3 months ago

C. Use the callback URL to disconnect the client from the backend service

This is essential for removing the client. Without it, the backend has no way to forcibly disconnect a problematic client.
✅ Must-have.
D. Add code to track the client status in Amazon ElastiCache in the backend service

Tracking client connection/disconnection behavior is critical to identifying problematic clients. ElastiCache is a highly suitable tool for storing connection data efficiently.
✅ Must-have.
E. Implement $connect and $disconnect routes in the backend service

While these routes are necessary for detecting client connections and disconnections, this could be seen as implicit in the implementation of D (since tracking client status assumes these routes are already used).
⚠️ Optional, depending on how the question is interpreted.

upvoted 1 times

Saurabh04 1 year, 8 months ago

Option DE: Implement $connect and $disconnect Routes:
Add $connect and $disconnect routes to your WebSocket API.
These routes handle client connections and disconnections.
When a client connects ($connect), a Lambda function can add the connection ID to a data store (e.g., DynamoDB).
When a client disconnects ($disconnect), another Lambda function can remove the connection ID from the data store.
Track Client Status:
Use Amazon ElastiCache (e.g., Redis) to track client status.
Store relevant information (e.g., client IDs, connection timestamps) in ElastiCache.
This allows you to identify clients that connect and disconnect

upvoted 1 times

9d8dd9c 1 year, 5 months ago

What about "the ability to remove the client"? for that it needs to use the callback function

upvoted 1 times

tomchandler077 1 year, 8 months ago

Option D ---> CORRECT Because, tracking the client's connection status using ElastiCache could help in identifying clients with erratic connection patterns and managing stateful information in a distributed environment, which is useful for WebSocket applications.

Also Option E CORRECT. These routes handle connection and disconnection events.

upvoted 1 times

tsangckl 1 year, 9 months ago

This appear at 17 Jun exam

upvoted 2 times

frangesk 1 year, 7 months ago

Do you remember the answers?

upvoted 1 times

65703c1 1 year, 10 months ago

Selected Answer: DE

DE is the correct answer.

upvoted 1 times

41eb566 2 years ago

Selected Answer: DE

E. Implement $connect and $disconnect routes in the backend service.

By implementing $connect and $disconnect routes, the backend service can capture when clients connect and disconnect from the WebSocket connection. This allows the application to track client status effectively.
D. Add code to track the client status in Amazon ElastiCache in the backend service.

upvoted 1 times

maurice2005 2 years ago

Selected Answer: CE

Your backend service can use the following WebSocket connection HTTP requests to send a callback message to a connected client, get connection information, or disconnect the client

https://docs.aws.amazon.com/apigateway/latest/developerguide/apigateway-how-to-call-websocket-api-connections.html

upvoted 1 times

SerialiDr 2 years, 1 month ago

Selected Answer: DE

When a client connects to your WebSocket API, the $connect route is invoked, and when they disconnect, the $disconnect route is invoked. You can use these routes to track the state of each client. By maintaining a record of each client's connections and disconnections, possibly in a database or an in-memory data store like Amazon ElastiCache, you can identify clients that frequently connect and disconnect.

Hence, the combination of changes that should be made to the application to meet these requirements includes:

Implement $connect and $disconnect routes in the backend service (Option E).
Add code to track the client status in Amazon ElastiCache in the backend service (Option D).

upvoted 2 times

KarBiswa 2 years, 1 month ago

Selected Answer: CE

C option - Supports https://docs.aws.amazon.com/apigateway/latest/developerguide/apigateway-how-to-call-websocket-api-connections.html
E option supports - https://docs.aws.amazon.com/apigateway/latest/developerguide/apigateway-websocket-api-route-keys-connect-disconnect.html

upvoted 1 times

Ashwinvdm22 2 years, 2 months ago

Selected Answer: CD

C: https://docs.aws.amazon.com/ko_kr/apigateway/latest/developerguide/apigateway-how-to-call-websocket-api-connections.html

D: You need a way to track which user is continuously reconnecting. That is why option D is so important because without it you will just be disconnecting every user that tries to connect cause then how will you know which user is the "problem" user. Note that you don't need the $disconnect endpoint to disconnect a client if you use option C. So CD is the only combination to solve the problem.

upvoted 2 times

Abdullah22 2 years, 2 months ago

going with DE

upvoted 1 times